MA342 Project 4

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1 Introduction

The ability to quantify uncertainty is a cornerstone of modern computational science. In financial markets, where price movements are dictated by a combination of trends and random fluctuations, mathematical modeling provides a framework for making decisions under risk. This report details a two-stage computational approach to market analysis based on the methodologies in Chapter 12 of Holder and Eichholz [1]. First, we employ stochastic differential equations to simulate potential future price paths for individual assets. Second, we utilize quadratic programming to construct an optimized portfolio that minimizes risk for a specific target return. By combining these methods, we can generate a predictive model for future wealth and evaluate its reliability using bootstrap statistical analysis.

2 Stochastic Simulation of Asset Prices

To model the behavior of a stock price over time, we assume that the price follows a Geometric Brownian Motion. This model is chosen because it ensures that stock prices remain positive and accounts for the fact that higher-priced stocks tend to have larger absolute price swings.

2.1 Mathematical Foundation

The continuous-time model for a stock price P is given by the stochastic differential equation:

$$dP = \mu P dt + \sigma P dz. \tag{1}$$

In this expression, μ represents the drift coefficient, which is the expected rate of return based on historical data. The parameter σ represents the volatility, or the standard deviation of the stock's returns. The term dz represents a Wiener process, which is a continuous-time random walk where the change over a small time increment dt is $dz = \epsilon\sqrt{dt}$, with ϵ being a random variable drawn from a standard normal distribution (mean of 0 and variance of 1).

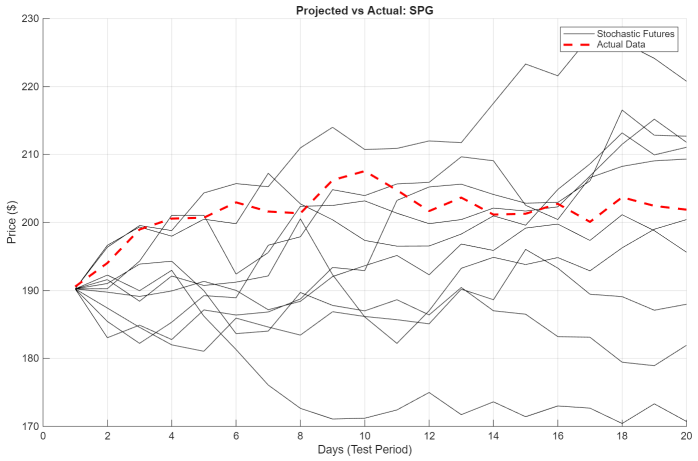


Figure 1: Monte Carlo simulation for SPG stock data.

To implement this in the model, we must find a discrete solution. Using Itō's Lemma, we can solve the differential equation for the discrete update formula used in the model:

$$P_{t+\Delta t} = P_t e^{\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\phi}. \tag{2}$$

In this equation, Δt is the time step (set to one day), and ϕ is a random value sampled from a standard normal distribution $N(0, 1)$. This formula allows us to simulate many possible futures for a stock, whence creating the Monte Carlo simulation. An example of a Monte Carlo simulation with 10 predictions compared to the actual data is shown in Figure 1.

3 Portfolio Optimization

While simulating individual stocks is useful, investors typically seek to reduce exposure (do u mean risk?) by holding a collection of assets. Portfolio optimization is the process of selecting weights for these assets to achieve the best possible risk-to-reward profile based on historical performance.

3.1 Defining Risk and Covariance

In this model, risk is defined as the variance of the portfolio's returns. To calculate this, we first determine the covariance between all assets in the selected subset. The covariance was found by calculating the covariance matrix, denoted as C, using the MATLAB function cov() on the historical daily returns of the assets. The risk of an individual asset, denoted as σ, is found by taking the square root of that asset's covariance with itself (C_{ii}), which is mathematically equivalent to the standard deviation of the asset's returns. By using the full covariance matrix, the model accounts for correlations amongst stocks, allowing

the optimization process to leverage diversification to lower overall portfolio variance.

3.2 The Quadratic Optimization Model

The model seeks to find a vector of weights w, where each element w_i represents the percentage of total capital invested in stock i. Following the framework in Holder and Eichholz [1], we formulate this as a quadratic (convex optimization) programming problem. The objective is to minimize the total weighted variance of the portfolio:

$$\text{Minimize: } \sum_{i,j} w_i C_{ij} w_j = w^T C w. \tag{3}$$

This optimization is governed by a set of linear constraints that ensure the resulting portfolio is both realistic and diversified. The first constraint is the full investment requirement:

$$\sum_{i=1}^{n} w_i = 1. \tag{4}$$

This ensures that 100% of the available capital is allocated across the selected assets. To prevent the model from over-concentrating funds into a single high-performing stock, an upper bound constraint is enforced such that $w_i \leq 0.20$ for all i. This requires the model to spread the investment across at least five different assets.The balance between risk and reward is controlled by the parameter α, which represents the user's risk tolerance on a scale of 0 to 1. We define r as the vector of expected returns for each asset. In this model, α acts as a weighting factor that determines the minimum acceptable expected return for the portfolio. By solving this system across the full range of α values, we generate the "Efficient Frontier" (Figure 2), which visualizes the optimal return possible for every level of risk.

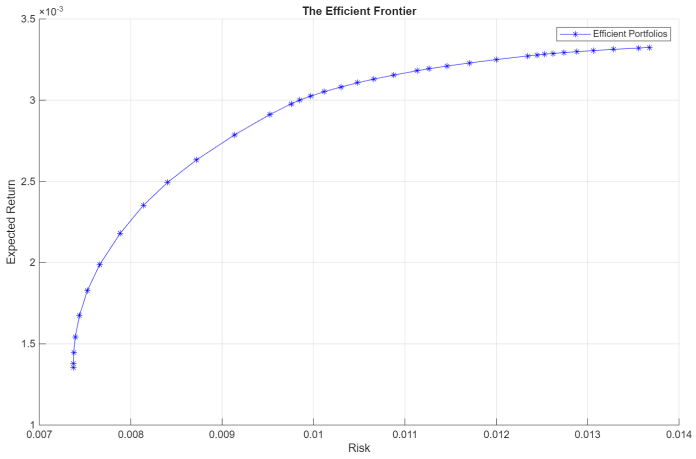


Figure 2: The efficient frontier for a select subset of stocks.

4 Predictive Results and Bootstrap Validation

To evaluate the performance of the optimized portfolios, the model combines the calculated weights with the stochastic simulations described in Section 2.

4.1 Bootstrap Confidence Intervals

Because a single stochastic trajectory represents only one possible outcome, a bootstrap method is used to verify the statistical significance of the results. This process is conducted across three distinct levels of risk (α values) to observe how the optimization constraints affect the final wealth distribution.

For each α, the model takes the final wealth outcomes from 100 simulations and resamples them with replacement 1,000 times. The mean of each resample is recorded, and the 2.5th and 97.5th percentiles are calculated to establish a 95% confidence interval. The resulting visualization (Figure 3) utilizes three subplots to compare these simulations based on a hypothetical \$1 initial investment over a period of 120 trading days. Table 1 then shows the calculated confidence intervals from the bootstrapping.

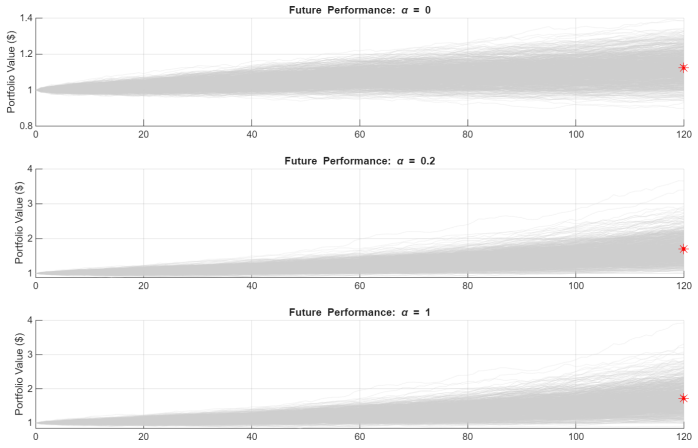


Figure 3: Visualization of simulations for three separate risk values.

Table 1: Bootstrapped Expected Portfolio Value and 95% Confidence Intervals Across Select Risk Tolerances (α)

α	**Expected Average Portfolio Value**	**95% Confidence Interval**
0.0	\$1.13	[\$1.12, \$1.13]
0.2	\$1.70	[\$1.68, \$1.72]
1.0	\$1.73	[\$1.70, \$1.75]

An important observation can be made from the data here in table 1 and the shape of the efficient frontier in figure 3. As α increases early on, the expected return increases greatly for little increased risk. This is most obvious when looking at the average portfolio return values. For just a 0.2 increase in risk, the expected returns skyrocket by 150%. Meanwhile, an addition increase in risk by 0.8 (a monumental step in comparison), only increases expected returns by 1%. This shows the importance of balancing risk vs reward in the portfolios as values at the extremes are often poor solutions.

4.2 Evaluation of Model Accuracy

When comparing the simulated trajectories to actual market data over a 20 day evaluation period, the limitations of the model become apparent. The actual market movement often falls outside the predicted confidence intervals. This highlight's a flaw in assuming that historical drift (μ) and volatility (σ) remain constant. In many test cases, the model predicts a steady gain while the actual data shows a significant loss, suggesting that the model lacks the complexity to account for sudden shifts in market sentiment.

The following figures (4 through 9) show the performance of the model compared against historical data. Five evenly spaced alpha values were chosen to see a spread of risk tolerances. All simulations are done with a hypothetical \$1

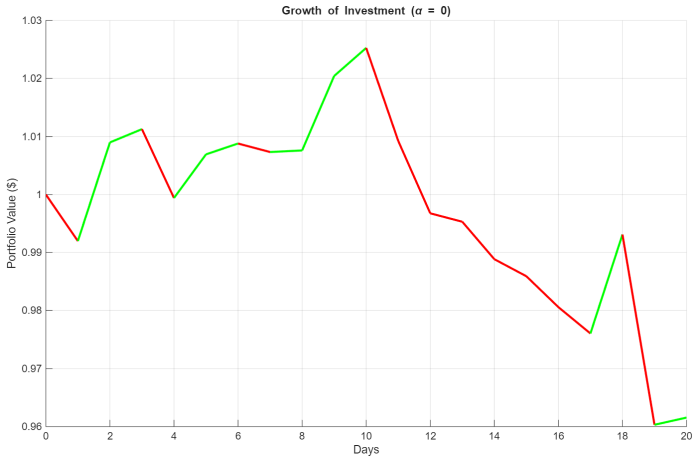


Figure 4: Portfolio evaluation with $\alpha = 0$.

initial investment and a table of final portfolio values can be seen below in table 2. It can also be noted that figures 8 and 9 are exactly identical because the model chose to invest in the exact same stocks for both risk values. Figures 6 and 7 may appear to be identical in results but the model chose slightly different weights for investing at these risk values.

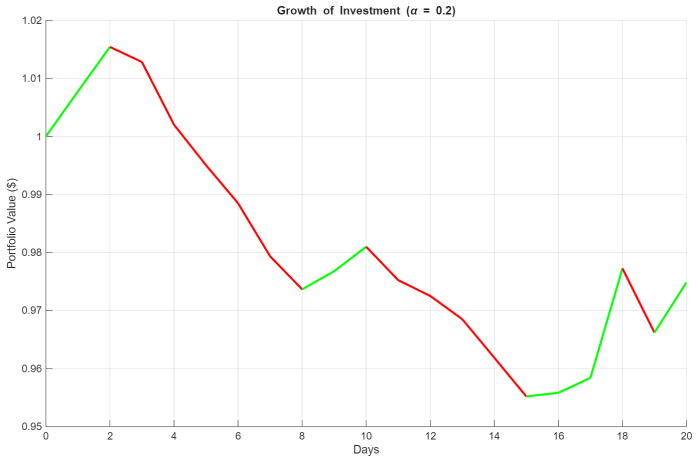


Figure 5: Portfolio evaluation with $\alpha = 0.2$.

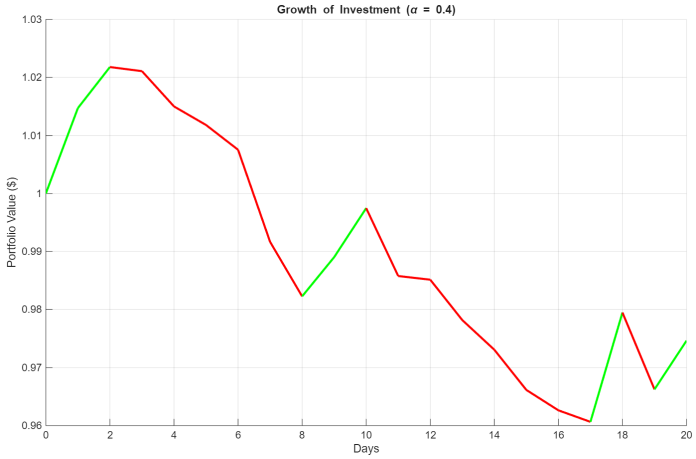


Figure 6: Portfolio evaluation with $\alpha = 0.4$.

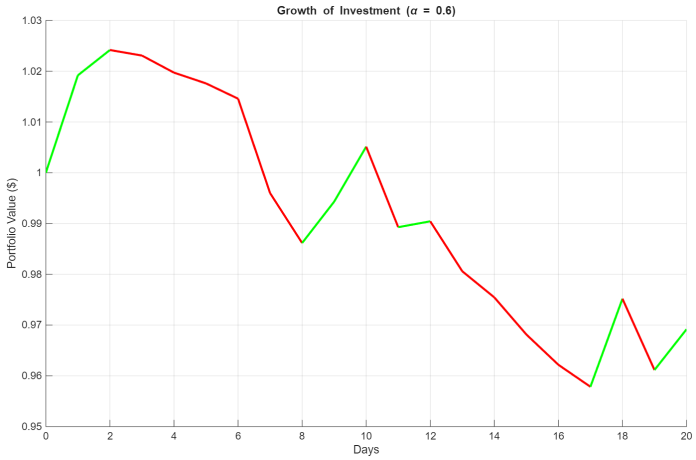


Figure 7: Portfolio evaluation with $\alpha = 0.6$.

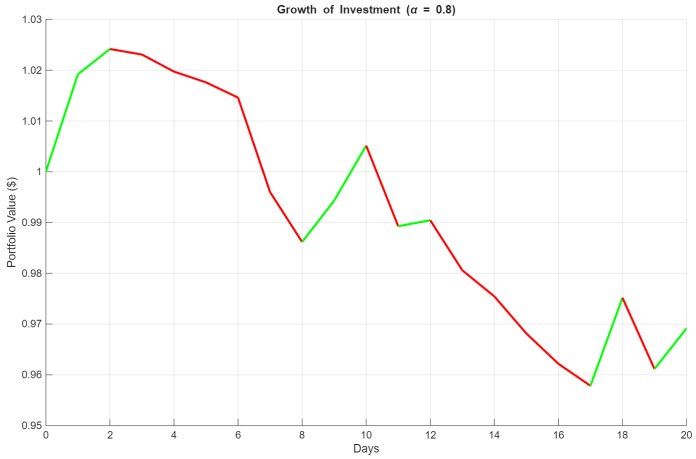


Figure 8: Portfolio evaluation with $\alpha = 0.8$.



Figure 9: Portfolio evaluation with $\alpha = 1$.

Table 2: Portfolio Values after 20 days for different risks

α	**Portfolio Value**
0.0	\$0.96
0.2	\$0.97
0.4	\$0.97
0.6	\$0.97
0.8	\$0.97
1.0	\$0.97

This data very clearly shows that the model is not accurate at predicting stock prices short term. With α values over the entire range, the model consistently loses money. This is not the guaranteed outcome of this model for all stock inputs and time frames, this is merely the results of the specific stock subset used and the time frame. For these figures, the model was run with the stocks found in the appendix over a time period of 05/04/2021 through 04/04/2026. The comparison of predicted vs actual was then done over the period of 04/04/2026 through 05/04/2026.

5 Conclusion

This study demonstrated the application of stochastic predictions and quadratic optimization to financial modeling. While the optimization successfully identifies portfolios that were mathematically optimal based on historical data, the discrepancy between the simulated results and actual performance suggests that historical data is a poor predictor of short term future movement. The model serves as a baseline for understanding risk vs return tradeoffs but requires more sophisticated understanding of how the market works to be considered an ef-

fective predictive tool. A representative image showing the mental state of the model's creators and authors of this report can be seen in the following figure, figure 10.



Figure 10: Summary of results.

References

[1] Allen Holder and Joseph Eichholz. *An Introduction to Computational Science*, volume 278. Springer Nature Switzerland AG, 2019.

Stock Ticker	**Stock Name**
HD	Home Depot
GE	GE Aerospace
MCD	McDonalds
COST	Costco
LMT	Lockheed Martin
RBLX	Roblox
SMCI	Super Mirco Computers Inc
AMD	AMD
MI	NFT Limited
BEN	Franklin Resources
SBR	Sabine Royalty Trust
CHH	Choice Hotels
SPG	Simon Property Group
NTAP	Netapp
FTDR	Frontdoor
AMGN	Amgen
CQP	Cheniere Energy
ITW	Illinois Tool
AMTU	Atmus Filtration
HESM	Hess Midstream
BLBD	Blue Bird
LAMR	Lamar Advertising
KSPI	Kaspi
BKE	Buckle
WDOFF	Wesdome Gold Mines

Table 3: Stonks used in the model